

Outliers Analysis

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Updated Evaluations Based on modified SINQ Llada

4-bit quantization of Dream-7B-Base and LLaDA-8B-Base. The numbers indicate the % of correct and incorrect answers of the baseline model that changed.

Model	ARC-Challenge	ARC-Easy	PIQA	Winogrande	MMLU	HellaSwag
LLaDA-8B-Base (SINQ)	3.58	3.49	1.69	6.47	8.60	1.65
LLaDA-8B-Base (AWQ)	3.75	3.24	1.58	6.08	8.80	1.18
LLaDA-8B-Base (GPTQ)	4.70	4.04	2.50	6.24	7.23	1.63
Dream-7B-Base (SINQ)	6.23	3.41	3.37	5.68	—	—
Dream-7B-Base (AWQ)	6.23	3.24	2.39	5.13	—	—
Dream-7B-Base (GPTQ)	5.20	3.33	2.61	5.84	—	—

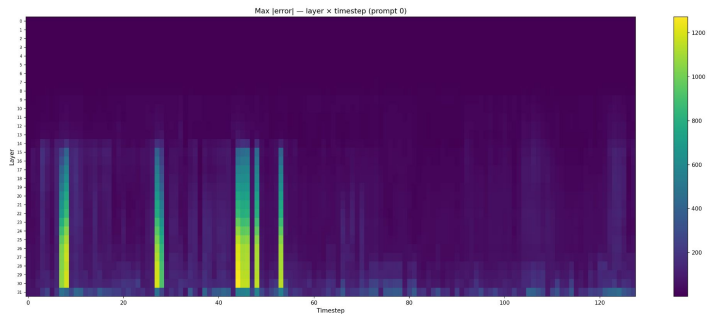
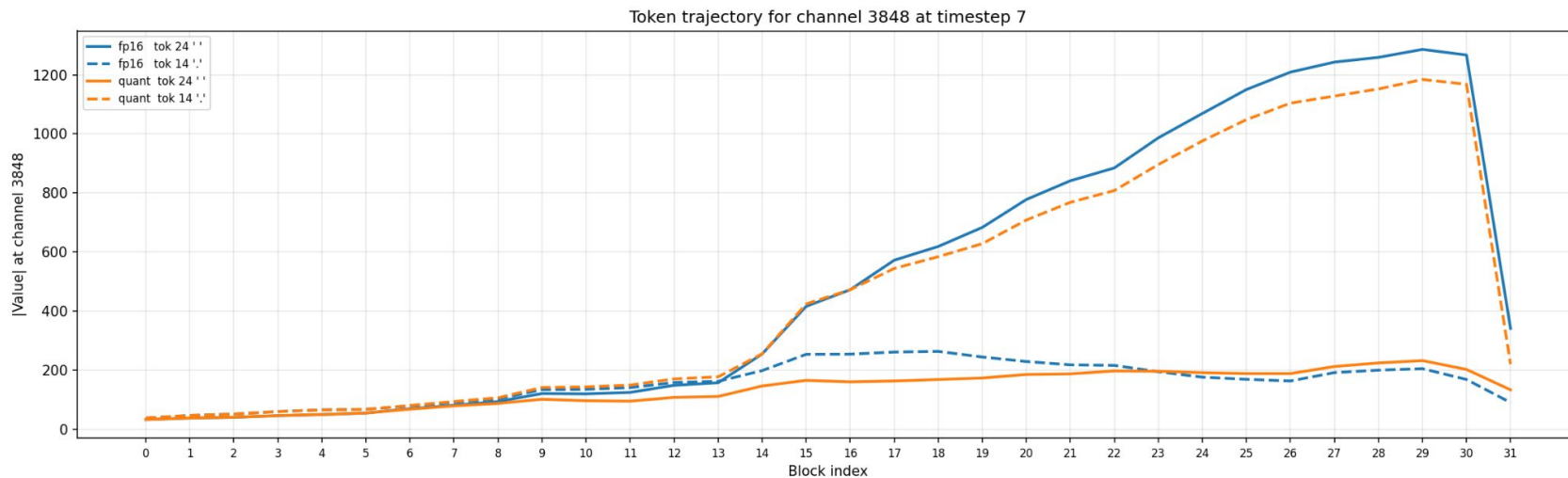
Updated Evaluations Based on modified SING Llada

Model	ARC-Challenge	ARC-Easy	PIQA	Winogrande	MMLU	HellaSwag
LLaDA-8B-Base (SING W4A16)	3.58	3.49	1.69	6.47	8.60	1.65
LLaDA-8B-Base (SING W4A8)	6.06	4.33	2.72	8.92	11.8	1.57
LLaDA-8B-Base (SING W4A4)	34.72	44.36	29.87	47.04	51.69	28.67
Dream-7B-Base (SING W4A16)	6.23	3.41	3.37	5.68	—	—
Dream-7B-Base (SING W4A8)	6.31	3.96	3.54	7.97	—	—
Dream-7B-Base (SING W4A4)	46.33	60.86	33.08	37.97	—	—

Outliers Analysis

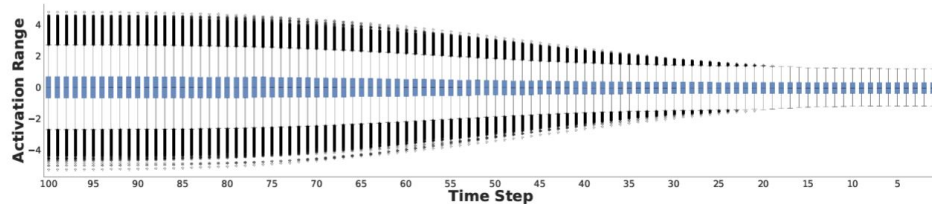
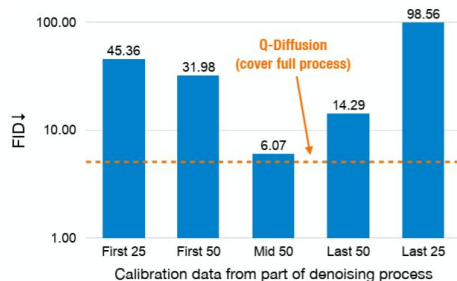
- In the llada model, massive outliers appear at the same channel throughout the layers at each timestep, channel 3848
- The outliers propagate through the model layers and keep increasing in magnitude.
- Max absolute error increases between quantized model and fp16 model when they disagree on which token position gets assigned with this massive outlier

Disagreement on Massive Outlier between fp16 and W4A16



Quantization Of Vision Diffusion Models - Q-Diffusion

- Discussed challenges in quantization due to:
 - Multistep inference
 - Accumulation of error across timesteps
 - Variation of activation distributions across timesteps



- Proposed solution: timestep-aware calibration sampling

Quantization Of Vision Diffusion Models - Q-Diffusion

- UNet Architecture
 - Analyzed exactly where large input activation ranges are appearing in the model

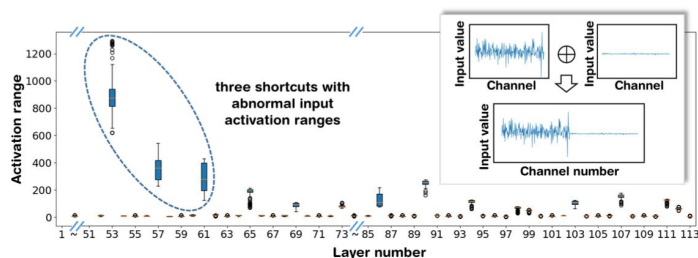


Figure 6: Activation ranges of DDIM's FP32 outputs across layers averaging among all time steps. We point out three shortcuts with the largest input activation ranges compared to other neighboring layers. Figures in the dashed box illustrate concatenation along channels. $\oplus$ denotes the concatenation operation.

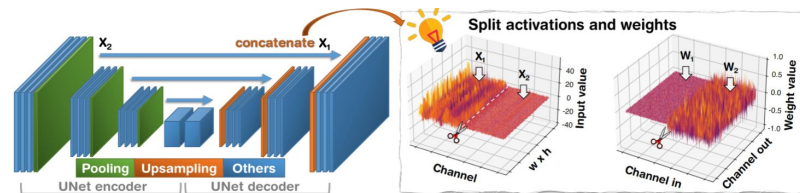


Figure 7: (Left) The typical UNet architecture with shortcut layers that concatenate features from the deep and shallow layers. (Right) The ranges of activations from the deep (X_1) and shallow (X_2) feature channels vary significantly, which also results in a bimodal weight distribution in the corresponding channels.

- Proposed solution is shortcut splitting quantization:
 - Performing quantization prior to concatenation in shortcut layers

Quantization Of Vision Diffusion Models - PTQD

- Using step aware mixed precision approach that aims to keep the SNR with reasonable range across all steps. This is done by adjusting the bitwidth of the activations

Table E: Bitwidth allocation for each dataset.

Dataset	W4A4 Step Range	W4A8 Step Range
ImageNet (250 steps)	249 to 202	201 to 0
ImageNet (20 steps)	19 to 15	14 to 0
LSUN-Bedrooms	199 to 155	154 to 0
LSUN-Churches	199 to 146	145 to 0

Quantization Of Vision Diffusion Models - PTQD

- Using step aware mixed precision approach that aims to keep the SNR with reasonable range across all steps. This is done by adjusting the bitwidth of the activations
- Dividing quantization noise to correlated and uncorrelated parts relative to the fully precise model

$$\begin{aligned}\mathbf{x}_{t-1} &= \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \hat{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z} \\ &= \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} (\epsilon_\theta(\mathbf{x}_t, t) + \Delta_{\epsilon_\theta(\mathbf{x}_t, t)}) \right) + \sigma_t \mathbf{z}.\end{aligned}$$

Quantization Of Vision Diffusion Models - PTQD

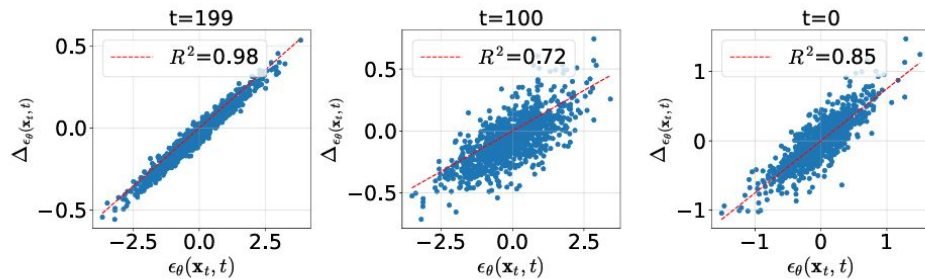


Figure 2: The correlation between the quantization noise (Y-axis) and the output of the full-precision noise prediction network (X-axis). Each data point on the plot corresponds to specific entries within these vectors. Data were collected by generating samples with 4-bit LDM-8 [49] for 200 steps on LSUN-Churches [65].

$$\Delta_{\epsilon_{\theta}}(\mathbf{x}_t, t) = k\epsilon_{\theta}(\mathbf{x}_t, t) + \Delta'_{\epsilon_{\theta}}(\mathbf{x}_t, t).$$

Quantization Of Vision Diffusion Models - PTQD

The correlated noise is corrected by dividing the output of quantized prediction model by $k+1$

Uncorrelated noise is corrected by calibrating the variance schedule and estimating the uncorrelated noise as gaussian distribution

$$\begin{aligned}\mathbf{x}_{t-1} &= \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} (\epsilon_{\theta}(\mathbf{x}_t, t) + \Delta_{\epsilon_{\theta}(\mathbf{x}_t, t)}) \right) + \sigma_t \mathbf{z} \\ &= \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \left((1 + k) \epsilon_{\theta}(\mathbf{x}_t, t) + \Delta'_{\epsilon_{\theta}(\mathbf{x}_t, t)} \right) \right) + \sigma_t \mathbf{z}.\end{aligned}$$

$$\Delta'_{\epsilon_{\theta}(\mathbf{x}_t, t)} \sim \mathcal{N}(\mu_q, \sigma_q).$$

Quantization Error Formulation for MDLMs

The conditional distribution for the reverse process for masked diffusion language models, for $0 \leq s < t \leq 1$, is factorized as:

$$q_{s|t}(x_s|x_t) = \prod_{i=1}^L q_{s|t}(x_s^i|x_t)$$

where the conditional distribution for each token is:

$$q_{s|t}(x_s^i|x_t) = \begin{cases} 1, & x_t^i \neq \mathbf{M}, x_s^i = x_t^i, \\ \frac{s}{t}, & x_t^i = \mathbf{M}, x_s^i = \mathbf{M}, \\ \frac{t-s}{t} q_{0|t}(x_s^i|x_t), & x_t^i = \mathbf{M}, x_s^i \neq \mathbf{M}, \\ 0, & \text{otherwise.} \end{cases}$$

where $q_{0|t}(x_s^i|x_t)$ is the categorical distribution over discrete vocabulary predicted by the model

Quantization Error Formulation for MDLMs

Let quantization error be denoted as $\Delta^i(x_s^i)$. The quantized mask predictor is equal to:

$$\hat{q}_{0|t}(x_s^i|x_t) = q_{0|t}(x_s^i|x_t) + \Delta^i(x_s^i)$$

Quantization error only affects the tokens that will be unmasked at this timestep, so the quantized mask predictor for these tokens equal:

$$\hat{q}_{s|t}(x_s^i|x_t) = \frac{t-s}{t} (q_{0|t}(x_s^i|x_t) + \Delta^i(x_s^i)) = q_{s|t}(x_s^i|x_t) + \frac{t-s}{t} \Delta^i(x_s^i)$$

$$q_{s|t}(x_s^i|x_t) + \frac{t-s}{t} \Delta^i(x_s^i) = q_{s|t}(x_s^i|x_t) \left(1 + \frac{\frac{t-s}{t} \Delta^i(x_s^i)}{q_{s|t}(x_s^i|x_t)} \right) = q_{s|t}(x_s^i|x_t) \left(1 + \frac{\Delta^i(x_s^i)}{q_{0|t}(x_s^i|x_t)} \right)$$

Quantization Error Formulation for MDLMs

By looking at the full sequence product instead of just a single token we get:

$$\hat{q}_{s|t}(x_s|x_t) = q_{s|t}(x_s|x_t) \prod_{i \in S_{unmask}} \left(1 + \frac{\Delta^i(x_s^i)}{q_{0|t}(x_s^i|x_t)} \right)$$

Let the full-precision logits be ℓ^i , such that $q_{0|t} = \text{Softmax}(\ell^i)$ and the quantized logit perturbation be $\delta\ell^i$ we can approximate softmax of the logits with perturbation using first order taylor expansion:

$$\text{softmax}(\ell + \delta\ell)_v \approx \text{softmax}(\ell)_v + \sum_{u=1}^{|V|} \frac{\partial \text{softmax}(\ell)_v}{\partial \ell_u} \cdot \delta\ell_u$$

After calculating the Jacobian of the softmax the result becomes:

$$\hat{q}_{0|t}(v) \approx q_{0|t}(v) + q_{0|t}(v) \cdot (\delta\ell_v^i - \bar{\delta\ell}^i)$$

Where $\bar{\delta\ell}^i = \sum_u q_{0|t}(u) \delta\ell_u^i$ is the probability-weighted expected logit shift

$$\Delta^i(x_s^i) \approx q_{0|t}(x_s^i|x_t) \left(\delta\ell_{x_s^i}^i - \bar{\delta\ell}^i \right)$$

$$\frac{\Delta^i(x_s^i)}{q_{0|t}(x_s^i|x_t)} \approx \frac{q_{0|t}(x_s^i|x_t) \left(\delta\ell_{x_s^i}^i - \bar{\delta\ell}^i \right)}{q_{0|t}(x_s^i|x_t)} = \delta\ell_{x_s^i}^i - \bar{\delta\ell}^i$$

Final Equation

$$\hat{q}_{s|t}(x_s|x_t) = q_{s|t}(x_s|x_t) \prod_{i \in S_{unmask}} \left(1 + \frac{\Delta^i(x_s^i)}{q_{0|t}(x_s^i|x_t)} \right)$$

$$\hat{q}_{s|t}(x_s|x_t) \approx q_{s|t}(x_s|x_t) \left(1 + \sum_{i \in S_{unmask}} \left(\delta \ell_{x_s^i}^i - \bar{\delta} \ell^i \right) \right)$$

This equation shows that degradation is strictly a function of high-variance logit perturbation relative to the mean shift